Dotted Lines and the Reconceptualization of Quantity in Early Modern Editions of Euclid's *Elements*

Abstract

This paper examines the use of dotted lines in the diagrams of early modern printed editions of Euclid's *Elements* and argues that their changing roles are closely tied to shifting conceptions of quantity. Based on a corpus of sixty extant editions published before 1650, the study traces the distribution, functions, and transformations of dotted lines across different editions and editorial contexts. In the arithmetical sections of the *Elements*, dotted lines represent discrete units and instantiate specific numerical examples. However, their use encounters practical and conceptual constraints in cases involving large values or hypothetical constructions, leading editors to adopt alternative visual strategies. In later editions, dotted lines are increasingly supplemented or replaced by Hindu–Arabic numerals, supporting more abstract representations of numbers and the relations between them. In the geometrical books, dotted lines appear later and mark continuous quantities, serving a range of visual functions and becoming ubiquitous in the seventeenth century. Close analysis of Book X shows how different representational conventions are integrated within a single diagram. Methodologically, the paper combines close visual reading with a digital workflow for large-scale diagram extraction and classification, allowing a systematic analysis. The findings show that dotted lines function as a flexible visual resource whose changing uses reflect and support broader shifts in the conceptualization of quantity in early modern mathematics.

Keywords: Euclid, Elements, Diagram, Digital humanities, Quantity, Early modern

1. Introduction

In recent decades, historians and philosophers have paid increasing attention to the role of diagrams. Where earlier accounts tended to focus on the textual dimensions of mathematical and scientific works, nowadays diagrams are widely regarded as integral parts of their reasoning, circulation, and intellectual content, shaping how concepts are expressed, understood, and communicated.[1]

The early modern transmissions of Euclid's *Elements* offer an especially valuable case for examining how visual language participated in the formation of mathematical concepts.[2] The *Elements* is composed of thirteen books, each presenting a sequence of propositions built upon one another. Compiled around 300 BCE, it had a sustained and broad influence across cultures since antiquity. During the early modern period, it was translated dozens of times and printed in more than two hundred editions (Wardhaugh et al., 2020). It was considered an ancient and foundational mathematical text and circulated across Europe in a range of contexts, from highly practical applications to the intellectual milieus of learned Europe (Høyrup, 2019). During this period, major changes, such as the advent of print, shifting audiences, and evolving intellectual and pedagogical priorities, prompted adaptations to both the text and its diagrams, resulting in a significant variation across editions (Baldasso, 2013; Oosterhoff, 2018; Pantin, 2022; Axworthy, 2023; Schubring, 2025).

One recurring visual feature in the editions of Euclid's *Elements* is the use of dotted lines. In this study, dotted lines are understood as rows of evenly spaced dots forming straight segments within diagrams. In geometrical contexts, such lines appear only sporadically in sixteenth-century editions of the *Elements* but become increasingly common in the seventeenth century.

[1] An extensive body of scholarship examines diagrams in historical mathematical texts, for example Netz (1999); Manders (2008); Saito and Sidoli (2012); Mumma et al. (2012); Chemla (2012). Other work discusses scientific diagrams more generally, not only mathematical ones, for example Baigrie (1996); Jones and Galison (1998); Kaiser (2005); Priest et al. (2018), while further studies address diagrams as visual objects within broader visual and cultural histories, for example Pauwels (2006); Bender and Marrinan (2010); Faietti and Wolf (2015); Bredekamp (2019).

[2] In addition to the studies mentioned in footnote 1, see, for example, Saito (2018); De Young (2005); Hamami and Mumma (2013); Macbeth (2010) on diagrams in Euclid's *Elements*, and Nasifoglu (2021); Schubring (2025); Alexander-Skipnes (2017); Heeffer and Van Dyck (2010) on diagrams in early modern mathematical and scientific texts.

The dotted lines served several functions in the geometrical diagrams, such as marking auxiliary constructions or indicating impossible configurations.[3]

If, however, attention is turned to the arithmetical parts of the *Elements*, where diagrams illustrate numerical conceptions of quantity, dotted configurations appear earlier and take on a distinct role. In these cases, each dot represents a discrete unit composing a number, and dotted lines function as visual aggregations rather than as extensions in space. As in the geometrical case, these configurations are attested primarily in printed editions, while studies suggest that they were largely absent from the earlier manuscript traditions of the *Elements* (Saito and Sidoli, 2012; Lee, 2020). Across the sixteenth and seventeenth centuries, the uses of dotted lines in both arithmetical and geometrical contexts vary considerably from one edition to another and do not follow a uniform trajectory.

These diagrammatic practices intersect with early modern discussions of quantity, which was often understood, in many senses, as a foundational object of mathematical inquiry (Andersen and Bos, 2006, p. 696). Traditionally, geometry and arithmetic are distinguished by their objects: geometry concerns continuous magnitudes, while arithmetic deals with discrete multitudes. While the notion of quantity had never been fixed, early modern scholars increasingly reworked how quantity was conceptualized.[4] By the end of the seventeenth century, with the consolidation of symbolic algebra and analytic geometry, the introduction of calculus and the blurring of the classical distinction between arithmetic and geometry, the concept of quantity had acquired a new form. Thus, tracing how this notion was reconceptualized during the early modern period can offer deeper insight into the transformations that reshaped the mathematical landscape of the time.

This study therefore asks: **How did the evolving visual language of**

[3]Lee (2018, 2020) discuss dotted lines, alongside other visual features, in the manuscript and early print traditions of Euclid's *Elements*. These issues are mentioned in Schäffner (2005); Axworthy (2023) as well. See also the discussion about Pythagorean dot notation in Steensen and Johansen (2016).

[4]This process is exemplified in the works of figures such as Simon Stevin (1548–1620), François Viète (1540–1603), Pierre de Fermat (1601–1665) and René Descartes (1596–1650). Stevin's *Arithmétique* (1585) redefined numbers as legitimate means of dealing with continuous magnitude, while Viète's algebra operated directly on geometric magnitudes. Fermat and Descartes further advanced these developments through algebraic treatments of geometrical problems. For broader discussions, see Goldstein (2000); Neal (2002); Malet (2006); Katz (2009); Naets (2010); Corry (2015); De Risi (2016); Oaks (2018).

dotted lines in early modern editions of Euclid's *Elements* participate in the reconfiguration of the concept of quantity? While the textual transmission of the *Elements* has been extensively studied, its visual dimension has received less systematic attention, and the changing roles of dotted lines have not been thoroughly examined. By focusing on these diagrammatic practices, this paper explores how early modern diagrams both reflected and perhaps shaped shifting understandings of numerical and geometrical quantity in the early modern period.

The paper proceeds as follows. The next section outlines the methodological framework, which combines close historical analysis with digital humanities techniques to study the corpus of early printed editions of the *Elements*. The subsequent sections examine the various uses of dotted lines in sixteenth- and early seventeenth-century editions, tracing their changing yet distinct dual form (Section 3) in the arithmetical books (Section 4) and geometrical books (Section 5). The discussion then turns to Book X (Section 6), where arithmetical and geometrical notions of quantity are introduced in the same diagrams. This analysis suggests that the boundary between the two conceptions of quantity was both challenged and blurred in the diagrams, a tendency reinforced by the growing use of Hindu–Arabic numerals. The paper concludes by considering several factors that may account for these visual developments and briefly points beyond the Euclidean corpus to dotted configurations in other mathematical, technical and scientific contexts (Section 7).

2. Corpus, methods, and digital workflow

This study draws on a corpus of sixty extant printed editions of Euclid's *Elements* published before 1650. The list of editions was compiled using a recently published catalog, *Euclid's in Print* (Wardhaugh et al., 2020), that aims to record all known early modern editions.[5] To offer a representative view of the translations of the *Elements*, the analysis focuses primarily on the first available edition issued by each editor or translator active during the period. In some cases, later editions by the same editor were also considered

[5] Wardhaugh et al. (2020) takes a conservative approach to the classification of a book as a rendering of Euclid's *Elements*, which can be challenged. For the complete list of editions selected in this paper see Table 2.

in order to identify both variations and instances of continuity. A few additional mathematical treatises and textbooks were examined to situate the *Elements* within a broader landscape of mathematical instruction and publication in the sixteenth and early seventeenth centuries and are discussed briefly in the final section of the paper.[6]

To investigate these texts at scale, each edition underwent an automatic diagram extraction workflow using *CorDeep*, a machine-learning system designed to detect and extract visual elements from scanned early modern books (Büttner et al., 2022). Originally built to assist historians and librarians in identifying scientific diagrams, *CorDeep* visually analyzes scanned pages and isolates diagram-like regions from textual content. Although not trained specifically on Euclidean diagrams, it proved effective in extracting a wide range of figures from the printed *Elements*. This allowed diagrams from hundreds of pages per edition to be rapidly segmented and then manually verified. *CorDeep* thus enabled a first-pass classification of diagrams, and of dotted lines in particular, across different parts of the editions, helping to define focused subcorpora for further analysis.

From there, the study moved to close visual reading. Diagrams containing dotted lines were examined in their textual and visual context to determine the function the dots served, for example, as representations of discrete units standing for numbers, or as auxiliary or hypothetical constructions. The diagrams were compared across multiple editions to trace how the same proposition was represented visually. To situate the patterned uses of dotted lines more broadly, they were compared with diagrams extracted from selected additional printed texts from the period. All identified diagrams, together with metadata and classification notes, have been made openly available in an online repository, both to allow replication of this study and to support future work in the history of mathematics and digital humanities.[7]

[6] Compiling the set of exemplary editions discussed in Section 7 required examining several dozen books. The full list of surveyed material is available in the project database (Anonymous, 2026).

[7] Full technical details can be found in Anonymous (2026).

3. The dual function of dotted lines: Discrete multitudes and continuous magnitudes

Analysis of the diagrams in the *Elements* shows that dotted lines fulfill two distinct visual roles. The first reflects an arithmetical function in which dotted lines represent a discrete multitude, individual units grouped together to form a number. The second reflects a geometrical function in which dotted lines represent a continuous magnitude, a line marked as distinct or auxiliary to the main figure. An illustrative example can be seen in Billingsley and Dee's 1570 English edition of the *Elements*. This edition employs both types of dotted lines in different sections of the edition, making it especially useful for comparison (see Figure 1). In XII.17 (Book XII, Proposition 17), for instance, a dotted line serves an auxiliary role. The dots are small and closely spaced, such that the line retains its visual continuity. By contrast, in the same edition, in VII.2, dotted lines appear in an arithmetical sense. The distance between the dots is large and each dot is distinct. They do not form a continuous line, but rather the number of dots along each segment corresponds exactly to the quantity being represented: eight dots between A and B, and four between C and D. Here, this visual aid to the proposition represents a concrete example that requires counting each dot individually. The dots do not indicate an extension, but rather a collection of discrete units.

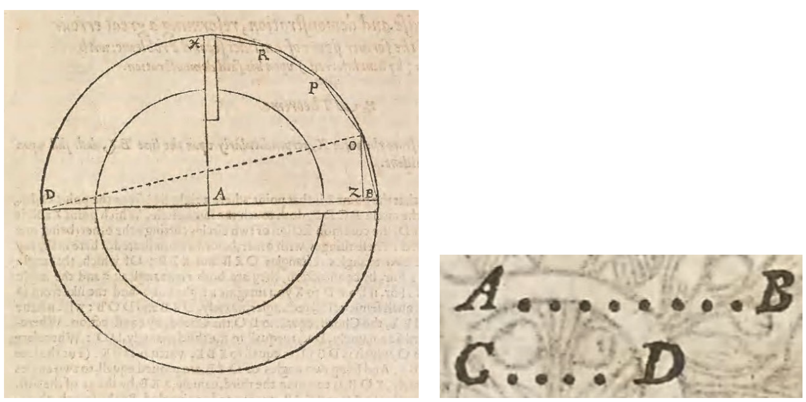


Figure 1: Geometrical and arithmetical uses of dotted lines in Billingsley and Dee's 1570 English *Elements*. The left diagram (XII.17) includes a dotted auxiliary line marking magnitude. The right diagram (VII.2) uses spaced dots to indicate numbers through discrete units.

The two types of dotted lines appear in separate sections of the examined editions of the *Elements*, reflecting the work's traditional division into arithmetical and geometrical books. The arithmetical books (VII–IX) include only diagrams that represent numerical quantities, and the arithmetical form of dotted line is confined almost entirely to them and to certain propositions in Book X, where results concerning numbers are used to explore incommensurability. The books that address plane (I–VI) and solid (XI–XIII) geometry, by contrast, do not include diagrams based on the discrete visual units. Rather, they employ dotted lines as representing continuous quantities with the continuous-style type of dotted lines.[8]

4. Dotted lines in Euclid's arithmetical books

4.1. Representing units as dots

The earliest visual use of a dotted line in the examined corpus appears in one of the earliest printed editions of the *Elements*, Lefèvre's 1516 edition. Dotted lines do not appear in the geometrical books of this edition, but in the arithmetical section, where they are used extensively in the arithmetical sense, representing discrete units. Proposition VII.9 offers an illustrative example (see Figure 2). This proposition states that if one number is a part of another, that is, it measures it a whole number of times, and a second pair of numbers stands in the same relation, then the ratio between the first number of each pair is equal to the ratio between the second terms.[9] The diagram in Lefèvre's edition presents this relationship through a concrete numerical example. The number 3 (labeled a) is composed of three dots, and the number 6 (b) of six dots, showing that the latter contains the former twice. A second pair, number 4 (c) and number 8 (d), is shown the same way. The visual structure of the dotted line allows the reader to "see" the

[8] A notable exception is Clavius's second edition (1589) and onward, which includes a few dotted lines in the arithmetic sense in the scholia to VI.17.

[9] The exact enunciation of the proposition in Lefèvre's edition reads: "Si fuerint quatuor numeri quorum primus secundi tota pars quota tertius quarti: erit permutatim to. ta pars aut partes primus tertij, quota pars aut partes secundus quarti." Heath's version of Euclid's proposition reads "If a number be a part of a number, and another be the same part of another, alternately also, whatever part or parts the first is of the third, the same part, or the same parts, will the second also be of the fourth" (Heath, 1908, vol. 2, p. 309). A parallel rendering in modern algebraic notation can be: if $a : b = c : d$, then $a : c = b : d$ (the permutability of ratios).

ratios by counting the units represented and to infer that the two numbers in the first line, number 6 (b) and number 8 (d), stand in the same ratio to the two numbers in the second line, number 3 (a) and number 4 (c).

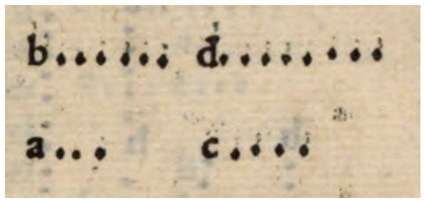


Figure 2: Dotted lines representing discrete quantities in the arithmetical section of Lefèvre's 1516 edition. Each dot corresponds to a unit, allowing visual comparison of the illustrated discrete quantities.

The concrete example is not spelled out in the text itself but supplied entirely by the diagram. This marks a noteworthy departure from the manuscript traditions. In them, not only are dotted lines absent, but even the solid lines used to represent numbers are typically of the same length and do not represent the metrical relations of the numbers they stand for (Saito, 2019). Saito proposes an explanation and argues that the lines are not meant to depict the numerical quantities themselves. Instead, their purpose is to illustrate the relationships and arrangements among the numbers involved in the argument and function as aids for memory and understanding. In this respect, both the appearance and the role of the manuscript diagrams differ from those in early printed editions.

However, the presentation of a specific example is not unique to the printed traditions of the *Elements*. Saito and Sidoli (2012) point out that the manuscript traditions also show a tendency toward overspecification, for instance by representing an arbitrary triangle as isoscele. They interpret this as part of the diagram's role as a schematic guide. They argue that in these cases the diagrams were meant to show some instantiation of the objects under discussion and act as anchors to the text, helping the reader track the argument and label positions, rather than illustrating all possible configurations. In this regard, the use of a specific example in the printed diagram can be understood, at least partially, as a continuation of earlier strategies of transmission, even as it also serves newer purposes, perhaps including pedagogical ones.

In the early printed editions, arithmetical diagrams with dotted lines instantiate the proposition by means of a concrete example. While the text sometimes speaks in general terms about "some" numbers, the diagram fixes specific values. The dots do not represent an abstract quantity. Rather, they mark a countable multitude that the reader can follow and enumerate. Some editions push this practice quite far, using very large numbers of dots, so that the example satisfies the proposition's conditions. For instance, in the diagram accompanying VIII.6 in Lefèvre's (1516) edition, repeated in Herlin's (1537) version, the segment labeled e contains no fewer than 81 dots (see Figure 3).

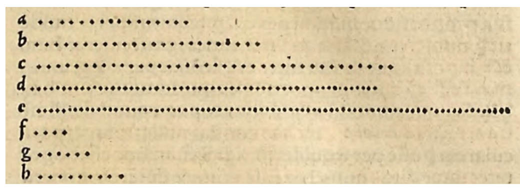


Figure 3: Dotted-line diagram for Proposition VIII.6 in Herlin's 1537 edition. Each dot represents a unit, illustrating the numerical relationships with the segment labeled e containing 81 dots.

This diagrammatic approach is not unique to Lefèvre's edition. Of the twenty-five translators that published versions of the arithmetical books and two that published editions that include only Book X, seventeen employ dotted lines at least once in their arithmetical diagrams. In every case where dotted lines appear, they are used in the same way as in Lefèvre's edition: each dot represents a single unit and the dotted line is lettered. In some of the cases, next to the dotted lines there are Hindu–Arabic numerals that specify the number of dots.

4.2. Challenges in dotted lines representation: Large numbers and impossible cases

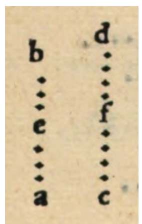
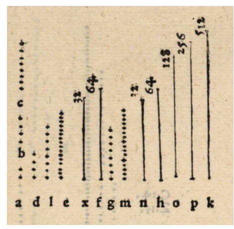
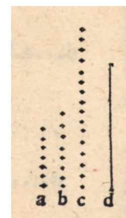
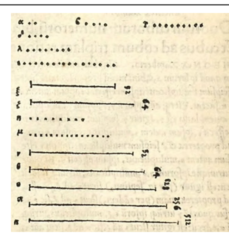
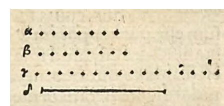
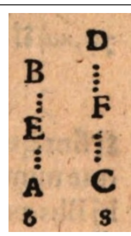
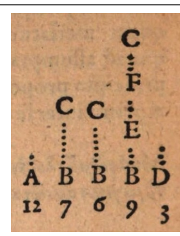
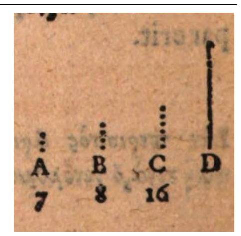
However, representing numbers through countable dots raised two main challenges, which were addressed in different ways across the corpus. First, some propositions involve numbers so large that a dotted line would need to contain an unwieldy number of dots. The second arose in *reductio ad absurdum* proofs, which depend on assuming the existence of a number that in fact does not exist, making it impossible to represent visually with any specific number of dots. The editions present a range of strategies for managing these difficulties. Their solutions not only respond to practical constraints, but also point toward, and perhaps even reinforce, a more abstract conception of number.

The various strategies for addressing large numerical values in diagrams are illustrated in the different visual treatments of VIII.13, which discusses continued proportion and therefore includes examples involving large numbers. These approaches are summarized in the second column of Table 1. A common method, used in the editions of Lefèvre (1516), Herlin (1537) and Commandino (1572; 1575), replaces the dotted line with a solid one and places a Hindu–Arabic numeral beside it to indicate its quantity. Magnien and Gracilis's edition (1564) adopts another solution: when confronted with large values, only a few dots (often two) accompanied by a Hindu–Arabic numeral are rendered. In many later editions, beginning with Billingsley and Dee (1570) and becoming common toward the end of the sixteenth century, dotted lines disappear altogether in such cases, leaving only Hindu–Arabic numerals with no additional visual marker.[10]

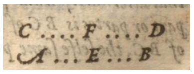
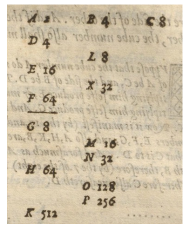
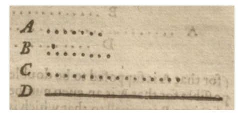
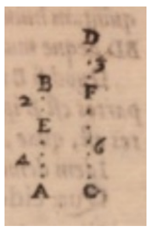
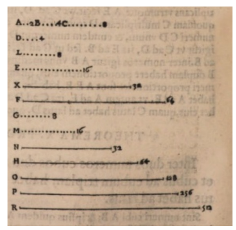
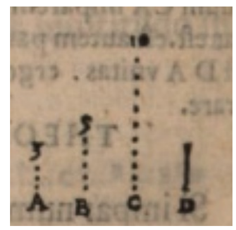
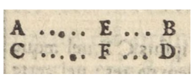
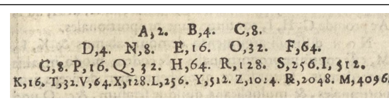
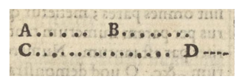
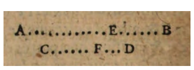
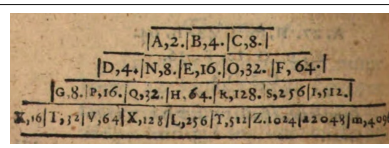
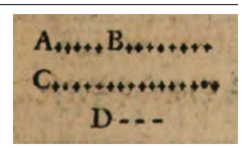
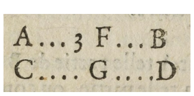
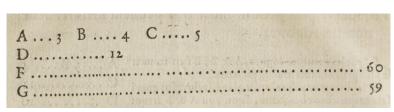
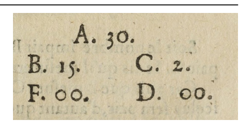
The second case that complicates the dotted-line convention in the arithmetical books appears in *reductio ad absurdum* proofs, which require assuming the existence of a number that actually violates the premises of the proof. Once again, editors adopted various solutions to visually signal the impossibility, as shown in the third column of Table 1. In the editions of Lefèvre (1516), Herlin (1537), and Commandino (1572; 1575), where large numbers are rendered as solid lines accompanied by Hindu–Arabic numerals, impossible numbers are indicated by solid lines devoid of numerals. The same strategy is also found in the editions of Magnien and Gracilis (1564), Billingsley and Dee (1570), and Clavius (1574, 1589, 1591). Three additional approaches were adopted in editions from the early seventeenth century. Rhode (1609) used a dashed line to mark the impossible case. Henrion (1615) employed a few dots, typically two, also without numerals. In his edition, dotted lines are ordinarily accompanied by Hindu–Arabic numerals, so their absence becomes a deliberate cue that the case is impossible. Dounot (1610) took a different route. While his edition includes dotted lines in Book VII, numbers are represented in the rest of the arithmetical books solely by Hindu–Arabic numerals, whether large or small. As a result, since Book VII contains no diagrams of impossible cases in his edition, this issue arises only in the diagrams of Books VIII–IX, where Dounot denoted impossible numbers with *00*.

[10]Billingsley's understanding of numbers is discussed in a broader context in Malet (2006).

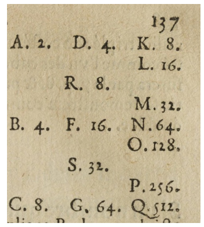
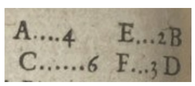
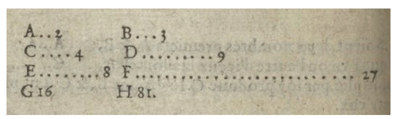
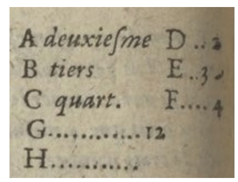
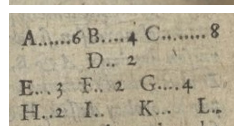
Table 1: Visual strategies for representing numbers in early printed Euclid editions.

Edition	Use of Hindu–Arabic numeral labels in diagrams	Adaptation for large numbers	Treatment of impossible (*reductio ad absurdum*) cases
Lefèvre (1516)	Never	Solid lines accompanied by Hindu–Arabic numerals	Solid lines
			
Herlin (1537)	Never	Solid lines accompanied by Hindu–Arabic numerals	Solid lines
			
Magnien and Gracilis (1564)	Always	A few dots (usually two) accompanied by Hindu–Arabic numerals	Solid lines
			
Billingsley and Dee (1570)	Never	Only Hindu–Arabic numerals next to the letter.	Solid lines

Continued on next page

Edition	Use of Hindu–Arabic numeral labels in diagrams	Adaptation for large numbers	Treatment of impossible (*reductio ad absurdum*) cases
			
Commandino (1572; 1575)	Always	Solid lines accompanied by Hindu–Arabic numerals	Solid lines
			
Clavius (1574, 1589)	Never	Hindu–Arabic numerals	Dashed lines
			
Rhode (1609)	Never	Hindu–Arabic numerals	Dashed lines
			
Dounot (1610)	Mostly	Book VII uses dotted lines and Books VIII–IX use Hindu–Arabic numerals	No such case in diagrams that include dotted lines, only in those with Hindu–Arabic numerals. There, the impossible case is represented with “00”.
			

Continued on next page

Edition	Use of Hindu–Arabic numeral labels in diagrams	Adaptation for large numbers	Treatment of impossible (*reductio ad absurdum*) cases
			
Henrion (1615)	Always	Hindu–Arabic numerals	Dotted lines accompanied by arbitrary number
			 

The table lists only those editions that contain at least one use of dotted lines in the arithmetical books. In every instance across all editions, the number of dots matches the numerical conditions of the proposition, and the dotted segment is always labeled with a letter. Unless otherwise noted, the examples in the table are taken from Proposition VII.11 (first column), VIII.13 (second column), and IX.31 (third column). Two sets of exceptions apply:

- *Adaptation for Large Numbers: VII.4 in the Magnien and Gracilis (1564) edition, VII.29 (corresponding to Heath (1908) VII.27) in Henrion (1615). In Dounot (1610), VII.38 (Heath, VII.36) is presented in addition to VIII.13, to illustrate the different treatment in Book VII and Books VIII–IX.*

- *Treatment of Impossible (Reductio ad Absurdum) Cases: IX.33 in Dounot (1610), and VII.41 (Heath, VII.39) and VII.35 (Heath, VII.33) in Henrion (1615).*

4.3. Exceptional cases in the arithmetical diagrams

A small number of exceptional cases stand apart from the general conventions established in each of the printed editions. One such case appears in the treatment of IX.9 in the editions of Lefèvre (1516) and Herlin (1537). The proposition states that if a sequence of numbers is in continued proportion, and the first term is a unit, then if the second is a square number, all the following ones are square and if it is cubic, they are all cubic.[11] In both editions, the diagram accompanying IX.9 consists solely of solid lines, without any Hindu–Arabic numerals. This choice breaks with the otherwise systematic use in these editions of solid lines with numerals to signify large, concrete numbers, and solid lines without numerals to mark impossible cases. Here, instead, the solid lines serve as an abstract placeholder for “any” number, detaching the diagram from specific numerical values and aligning it with the proposition’s generality. The effect is softened slightly by the diagram for IX.8, which immediately precedes it and shares the same visual structure but includes numerals beside the lines. In this sense, IX.8 functions as a cushion, preparing the reader for the diagram of IX.9. This is the sole instance in either edition where a diagram in the arithmetical books can be understood as conveying the notion of “any” number rather than a specific value.

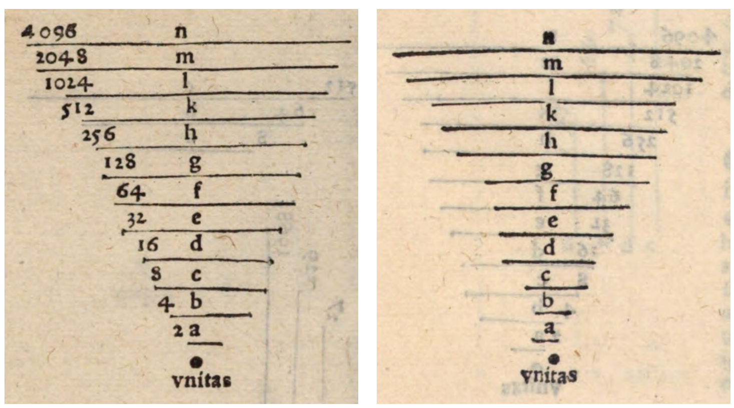


Figure 4: Diagrams for Propositions IX.8 and IX.9 in Lefèvre (1516)’s edition. While IX.8 includes numerals indicating concrete values, IX.9 replaces them with plain lines.

[11] In Euclid’s terminology, numbers are in continued proportion when each term has the same ratio to the next. In modern notation, this notion is often expressed as $a : b = b : c = c : d$ and so on. Proposition IX.9, shows that if the first term in such a sequence is a unit and the second is a square number (a product of a number by itself, like 4, 9, 16, etc.), all subsequent terms are also square and if it is cubic (a number taken three times in multiplication like 8, 27, 64, etc.), they are all cubic. The original in both editions, based on Campanus’s version, is "Si numeris quolibet ab unitate continua proportionalitate dispositis, unitatem sequens quadratus fuerit, caeteri quoque omnes erunt quadrati. Si vero qui unitatem sequitur fuerit cubus, caeteri quoque omnes erunt cubici."

Another exceptional case involves the appearance of diagrams in the definition section of Book VII. Out of the nine editors that use dotted lines in their rendering of Books VII–IX, four, Clavius (1574, 1589, 1591), Billingsley and Dee (1570), Rhode (1609) and Henrion (1632), include diagrams alongside certain definitions in Book VII. Two definitions are particularly noteworthy: VII.16, defining the plane number, and VII.17, defining the solid number. Since these are the arithmetical books, "plane" and "solid" refer to numbers, not magnitudes. According to Definition VII.16, a plane number is one that results from multiplying two numbers together. In Definition VII.15, multiplication in the numerical sense is itself defined as repeated addition: a number is multiplied when it is added to itself as many times as there are units in another number.

In both Clavius's and Billingsley and Dee's editions, the definitions for plane and solid numbers are accompanied by diagrams that illustrate the resulting numbers as rectangles or cubes. Rhode, by contrast, includes only the cubic form. Henrion (1632) includes only the rectangle form. Interestingly, his earlier editions, from 1615 and 1621, did not include a diagram for Book VII's definition. In his later edition, the diagram appeared as part of the commentary. It illustrates a specific example, accompanied by Hindu–Arabic numerals and a text that explains it. Such shapes are by no means required by the definition. For instance, multiplying 4 by 3 could simply be shown as adding a group of 4 three times. Yet the rectangular and cubic forms used by these editions may suggest an underlying geometric or at least spatial mode of thinking, a visual metaphor that subtly bridges arithmetic with spatial representation.[12]

In general, the later editions show a marked decline in the use of dots, with Hindu–Arabic numerals increasingly used in their place, even for small numbers. For instance, in Dounot (1610) dots are used consistently throughout Book VII, while Books VIII–IX rely exclusively on Hindu–Arabic numerals in the diagrams, without dotted-line representation. In a way, replacing dots with numerals in the diagrams reintroduces a characteristic of the manuscript tradition: the metrical relations between numbers are no longer immediately visible at a glance. Unlike dotted lines, numerals lack spatial indication and require the reader to interpret the values rather than perceive them visually.

[12] For some notes regarding uses of dot notation for triangular, oblong, and square numbers, see Steensen and Johansen (2016).

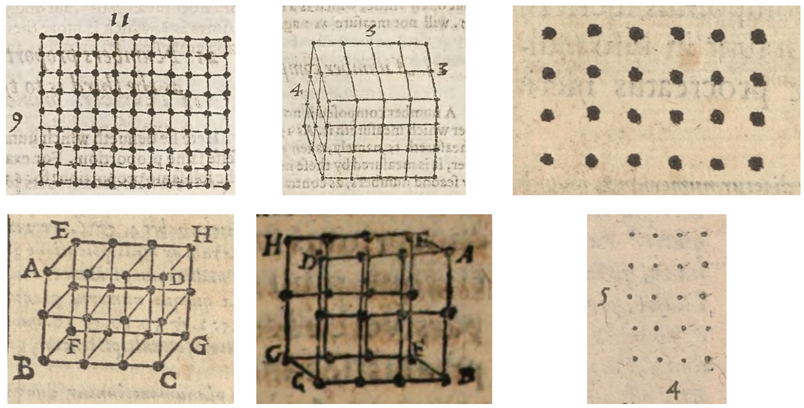


Figure 5: Diagrams illustrating the definitions of plane and solid numbers (Definitions VII.16–17). The diagrams are taken from four editions (clockwise from upper left): Billingsley and Dee (1570) (Definitions VII.16–17), Clavius (1574) (Definitions VII.16–17), Rhode (1609) (Definition VII.16) and Henrion (1632) (Definition VII.17). In these diagrams, numerical multiplication is visualized through rectangular or cubic configurations, translating arithmetical relations into spatial forms.

Yet unlike in the manuscripts, where the values were often left abstract, the diagrams in the printed editions that rely on Hindu–Arabic numerals provide concrete examples, anchoring the diagrams in specific numerical instances.

The use of Hindu–Arabic numerals in diagrams also opened new opportunities, which editors of the time were quick to seize. A particularly striking example is Clavius's second edition of the *Elements* (1589).[13] While the diagrams for Book VII's propositions remain unchanged from his first edition (1574), Clavius expands the scholia to illustrate how these propositions apply to fractions. Thus, even though the diagrams for the propositions themselves still use dotted lines, the scholia of the second edition include fractions represented in the same style that earlier editions reserved for large integers. The first example of this appears in VII.5. The proposition states that if one number is the same part of another and a second number is the same part of a third, then the sum of the first pair is the same part of the sum of the second pair. For instance, 6 is half of 12, and 4 is half of 8, so their sum, 10, is half of 20. In the scholia of this proposition, Clavius begins by declaring that

[13]Six decades later, in his 1632 edition, Henrion added a section titled "Demonstrations des fractions ou nombres rompus" to the end of Book VII. In this section, Henrion uses diagrams that include simple fractions notations as well.

the same is true for fractional numbers ("Idem hoc verum est in numeris fractis"), and proceeds to demonstrate it.[14]. While the diagram accompanying the main proposition uses dotted lines to represent the numbers, the parallel diagram in the scholia replaces these dots with fractional notation (see Figure 6). Notably, the structure and placement of letters remain identical in both diagrams. This mirroring effect extends to the demonstration itself, reinforcing rhetorically that the mathematical logic governing numbers in the traditional arithmetical sense applies equally to fractions.

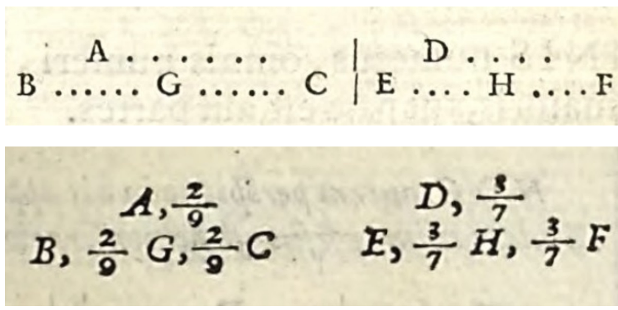


Figure 6: Diagrams accompanying Proposition VII.5 in Clavius (1589). The upper diagram represents numbers using dotted lines, while the lower, which appears in the scholia section, replaces the dots with fractional notation, extending the same visual logic to fractions.

5. Dotted lines in Euclid's geometrical books

While the use of dots in the arithmetical books is well established in the early editions and declines towards the end of the sixteenth century in favor of Hindu–Arabic numerals, dotted lines appear in the geometrical books only toward the end of the sixteenth century and become ubiquitous in the seventeenth century, spreading across a wide range of uses and numerous diagrams. Dotted lines served various purposes in the geometrical books: project unseen elements of a solid figure, indicate compass arcs, show that two lines are drawn parallel to one another, signal the hypothetical step

[14]For an additional analysis of Clavius's understanding of the concept of number, see also Malet (2006)

involved in a *reductio ad absurdum* argument, and more.[15] In all these cases, the same visual form of the dotted line is employed, regardless of function. This uniformity is evident, for example, in Rudd (1651), where dotted lines appear in I.5 to mark auxiliary lines added during the demonstration, and in I.8 to denote an impossible configuration introduced for contradiction (see Figure 7).

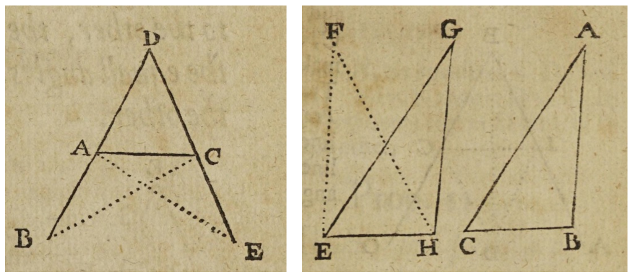


Figure 7: Usages of dotted lines to represent geometrical magnitudes. The diagrams accompany Propositions I.5 (left) and I.8 (right) in Rudd's 1651 edition of the *Elements*. The dotted lines mark auxiliary constructions and indicate hypothetical or impossible configurations introduced for contradiction correspondingly.

Scheubel's (1550) Latin and Xylander's (1562) German editions of the *Elements* are the earliest in the corpus under study to employ dotted lines extensively in the geometrical sense (see Table 2). Both contain only the first six books, which treat plane geometry. Their diagrams are independent and unique, neither copied from the other, with dotted lines fulfilling multiple functions. In I.7, for example, both depict the impossible configuration considered in the proof with dotted lines. The proposition claims that two different triangles cannot share the same base and lie on the same side of it while having all corresponding sides equal. As shown in Figure 8, each edition included impossible configurations using dotted lines to mark triangles that clearly do not share sides of the same magnitude as the main triangle. However, the examples chosen in each edition are different. For both editors the representation of absurd cases is one of the principal uses of dotted lines, but this tool also appears in other contexts as well. In Xylander's diagram for II.14, for example, dotted lines serve as auxiliary constructions that support the reasoning of the proof.

[15] Some purposes of dotted lines, mainly in the Latin translation, are discussed in (Lee, 2018).

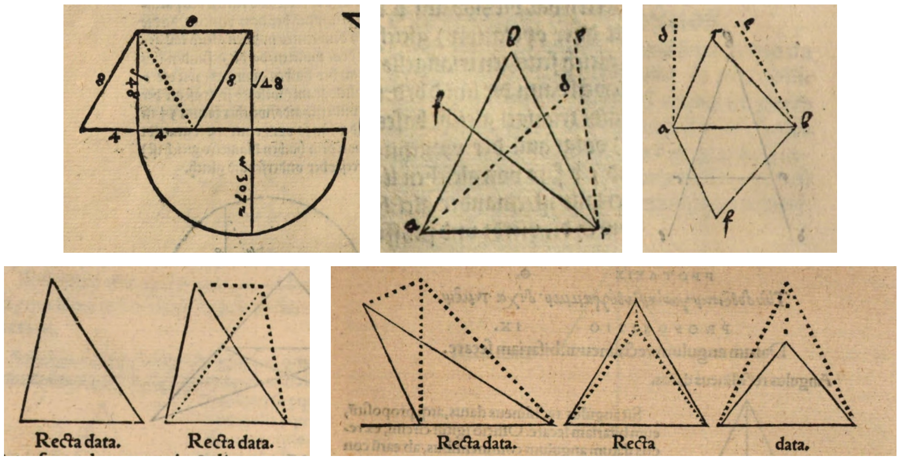


Figure 8: Diagrams with dotted lines from Xylander's 1562 and Scheubel's 1550 editions. The top row shows an example added by Xylander after II.14 (left) together with two diagrams for I.7 from the same edition. The bottom row presents Scheubel's diagram for the same proposition. Dotted lines mark impossible configurations (in I.7) and auxiliary constructions (in the added example), sometimes accompanied by concrete numerical values indicated with Hindu–Arabic numerals and radical notation ($\sqrt{\ }$).

The next edition in the corpus to use dotted lines extensively in the geometrical books is Clavius's second edition of the *Elements* (1589). Clavius, an esteemed mathematics professor at the Collegio Romano, exerted significant influence on mathematical education through both his pedagogical innovations and the wide circulation of his works and approaches. He was also one of the most influential translators of the *Elements*.[16] His Latin translation of Euclid's *Elements* appeared in at least nine editions between 1574 and 1655 in Rome, Frankfurt, and Graz, and he was frequently cited by others (Wardhaugh et al., 2020). Although his first edition (1574) introduced several diagrammatic features, such as shading and compass arcs, the use of dotted lines appears in it only once, in II.1. By contrast, in his second edition, dotted lines are used extensively, a pattern that continues in his subsequent editions.

Later, dotted lines became a common feature in editions of the *Elements*.

[16]For an additional discussion on Clavius's role in early modern mathematics and the Euclidean tradition, see Rommevaux (2006).

Within the examined corpus, among the twenty-eight translators who published editions of the geometrical books before 1600, only three employed dotted lines widely, and three used them sporadically in one or two instances. By contrast, of the nine translators who published between 1600 and 1650, five relied heavily on dotted lines and one used them in a small number of propositions. Moreover, for three translators, Henrion, Clavius and Dounot, the use of dotted lines increased between their first edition and later editions.

Table 2: Dotted lines in Euclid's *Elements* editions. Occurrence of dotted lines in arithmetical and geometrical books of Euclid's *Elements* across editions ordered by publication date. *Not applicable* cells indicate editions that do not include arithmetical books (second column) or geometrical books (third column). Empty cells indicate the absence of dotted-line diagrams in the corresponding books.

Edition	Edition includes diagrams	Dotted lines in arithmetical Books	Dotted lines in geometrical Books
Campano (1482)	V		
Zamberti (1505, 1510)	V		
Pacioli (1509)	V		
Lefèvre (1516)	V	V	
Grynaeus (1533)	V	V	
Fine (1536, 1544, 1551)	V	*Not applicable*	
Herlin (1537, 1546, 1559)	V	V	
Anonymous (1538)	V	*Not applicable*	
Tartaglia (1543, 1565, 1569, 1585, 1586)	V		
Caiani (1545b,a)			
Rhäticus and Camerarius (1549)	V	*Not applicable*	
De la Ramée (1549, 1558)			

Continued on next page

Edition	**Edition includes diagrams**	**Dotted lines in arithmetical Books**	**Dotted lines in geometrical Books**
Scheubel (1550)	V	*Not applicable*	Multiple diagrams in Books I, III and VI
De Montdoré (1551)	V	V[17]	
Scheubel (1555)	V		*Not applicable*
Naboth (1556)			
Peletier (1557)	V	*Not applicable*	
Anonymous (1559)	V	*Not applicable*	
Xylander (1562)	V	*Not applicable*	Multiple diagrams in Books I–IV and VI
Magnien and Gracilis (1564)	V	V	
Forcadel (1564)	V	*Not applicable*	I.43
Sthen (1564)			*Not applicable*
Forcadel (1565)			*Not applicable*
Forcadel (1566)	V	*Not applicable*	
De Foix-Candale (1566)	V		
Billingsley and Dee (1570)	V	V	VI.6, XII.17
Commandino (1572)	V	V	
Clavius (1574)	V	V	
Commandino and Spacciuoli (1575, 1619)	V	V	
Maurolico (1575)	V	*Not applicable*	
Zamorano (1576)	V	*Not applicable*	

Continued on next page

[17] The editions of De Montdoré (1551) and Du Puy (1612) contain only Book X. In both, dotted lines are employed in an arithmetical sense, appearing as arrays of dots where each dot denotes a single unit.

Edition	Edition includes diagrams	Dotted lines in arithmetical Books	Dotted lines in geometrical Books
Clavius (1589)	V	V	I.def.40, I.12, I.28, I.47, II.12–14, IV.5, IV.16, VI.11, VI.17
Clavius (1591)[18]	V	V	Multiple diagrams in Book I, II.12, II.13, III.11, III.12, IV.5, IV.11, VI.17, VI.33, XIII.13
Errard (1598)	V	*Not applicable*	III.25
Dyvad (1603)	V	*Not applicable*	
Dou (1606)	V	*Not applicable*	Multiple diagrams in Books I–IV and VI
Rhode (1609, 1634)	V	V	Multiple diagrams in Books I, III, IV, VI
Peletier (1610)	V	*Not applicable*	
Marius (1610)	V	*Not applicable*	Multiple diagrams in Books I, III and VI
Dounot (1610)	V	V	
Du Puy (1612)	V	V[17]	
Dounot (1613)	V	V	Multiple diagrams in Book I
Henrion (1615, 1621)	V	V	VI.18
Lanz (1617, 1629)	V	*Not applicable*	
Peletier (1628)	V	*Not applicable*	
Henrion (1632)	V	V	III.22, III.23, III.25, VI.18, VI.20
Gestrinius (1642)	V	*Not applicable*	Multiple diagrams in Book I, II.12, II.13, IV.def.3 and on the title page

[18] Clavius's 1591 edition also includes a section at the end of Book VI on the nature of curved lines, which uses dotted lines, in the geometrical sense, extensively.

Both Clavius's third edition (1591) and Rhode's edition (1609) offer a particularly striking case: the same visual form of the dotted line appears in both the arithmetical and geometrical senses (see Figure 9). In Clavius's first definition in Book V, the diagram shows a small segment A that measures two larger segments, B and C, marked with division lines indicating how many times A fits into them. Two additional segments, E and F, are not measured by A, and dots are added to extend their solid lines to show what would need to be added for A to measure them. In Rhode's edition, the impossible case in the arithmetical books is rendered using a dashed line, and the same dashed line is employed to represent auxiliary lines in the geometrical diagrams. There is no indication that either Clavius or Rhode intended to draw an explicit connection between the two uses, nor is such a connection especially meaningful in mathematical terms. A more plausible explanation is practical, perhaps related to the printing process.[19] Nevertheless, the result is noteworthy: a single visual vocabulary is applied across both arithmetical and geometrical representations of quantity.

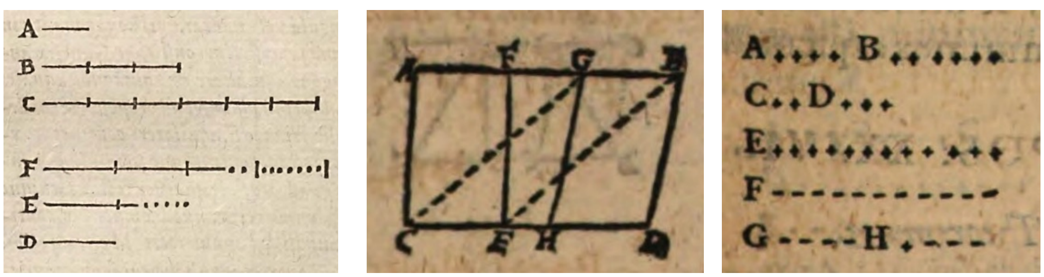


Figure 9: Uses of dotted lines in arithmetical and geometrical contexts. The left diagram is taken from Clavius's (1591) first definition in Book V, while the center and right diagrams correspond to Rhode's (1609) Proposition I.36 and VII.36, respectively. In these diagrams the dotted line employs the same visual form while operating in both arithmetical and geometrical senses.

6. Dotted lines in Book X

The reception of Book X of Euclid's *Elements* is an especially compelling case study for understanding shifts in the early modern conception of quantity. Book X deals with incommensurable magnitudes, quantities that cannot

[19] For an additional discussion about the practical considerations imposed by the printing process of diagrams, see Baldasso (2013); Pantin (2022).

be measured by a shared unit regardless of how many times it is applied. The notion of incommensurability in Book X is notoriously difficult, and the book itself is widely regarded as one of the most complex and challenging parts of the *Elements*. In the early modern period, as well as in other contexts, it prompted a wide range of inventive explanatory strategies and fostered experimentation with textual exposition, diagrams, and symbolic forms. The scholars of the time studied it extensively, offering abbreviated or adapted versions, for both didactical and intellectual reasons. In their hands, Book X became a bridge between traditional geometrical frameworks and emerging early modern conceptions of number, algebra, and abstraction. In this way, Book X helped lay the groundwork for the modern exploration of irrationality, with which it is often identified.[20]

Book X explicitly builds on propositions from the arithmetical books to advance its inquiry into incommensurability. The first clear articulation of this dependence appears in X.5, which states that commensurable magnitudes stand to one another in the same ratio as numbers do.[21] In this way, the proposition establishes a structural correspondence between relations between magnitudes and relations between numbers. The strategy it exemplifies recurs throughout the book, and some early modern thinkers understood it in the following way: questions about magnitudes are reconsidered in numerical terms, addressed using arithmetical propositions, and then restated within a geometrical framework.[22] Propositions X.5–9, along with the two lemmas preceding X.29, take the relationship between numbers and magnitudes as their very subject, articulating it explicitly within their enunciation.

In the examined editions, two distinct approaches emerge in the diagrams

[20]For an additional discussion on Book X, its early modern reception, and its nuanced connection to the concept of irrationality, see Fowler (1992); Artmann (1999); Corry (2015); Wagner and Netz (2023). The difficulty of Book X is also explicitly addressed in early modern editions, for example, in Billingsley and Dee (1570). On the debated arithmetical interpretation of Book X, see Wagner and Netz (2023). For a general discussion of "geometric algebra" see Høyrup (2017).

[21]The phrasing follows Heath (1908)'s translation, which can be challenged. There is also some analogy between Propositions X.2–4 and VII.1–3, see Heath's commentary on X.2 and Wagner and Netz (2023, p. 85–86). However, this resemblance is perhaps best understood as an analogy, see Mueller (1981, pp. 136–38). Saito (2019) discusses additional and more nuanced connections between Book X and Books VII–IX.

[22]See Billingsley and Dee (1570) as an example.

accompanying Proposition X.5, both mirroring the visual conventions used in the arithmetical books of the same volume. With the exception of De Montdoré (1551) and Du Puy (1612) editions, which include only Book X, every edition that uses dotted lines in Book X also uses them in Books VII–IX and preserves in the diagrams of Book X the visual vocabulary employed for units and numbers. Across all of the editions in the examined corpus, the diagram for X.5 presents six quantities: three magnitudes and three arithmetical quantities (two numbers and a unit), arranged in corresponding relation. The magnitudes are always rendered as three solid lines subdivided to indicate their commensurability.

In contrast, the numerical quantities are treated in two different ways (see Figure 11). One group of editions represents them with dotted segments, while the other replaces the dots with Hindu–Arabic numerals, a practice first attested in Clavius's 1574 edition. In some of the dotted-line editions, Hindu–Arabic numerals are added next to the dotted segment to specify explicitly how many units the dots stand for. The division between these strategies, and between those editions that label the dotted segments and those that do not, mirrors the diagrammatic conventions already in place in the arithmetical books. Thus, in editions that frequently employ Hindu–Arabic numerals in Books VII–IX, even for small values, the diagram for X.5 tends to rely entirely on them, and in those that add Hindu–Arabic numerals alongside dotted segments in the arithmetical books, that practice is carried over here as well.

Overall, the different diagrams for X.5 exemplify how the boundaries between the two concepts of quantity become less sharp. In these diagrams, the elements that represent numbers, whether dots or Hindu–Arabic numerals, do not simply coexist with their geometric counterparts, but echo them, forming a kind of symbolic mirror that underscores their shared relational structure. Moreover, Hindu–Arabic numerals begin to operate as more than a simple substitution for dots. They serve increasingly as an intermediary symbol, bridging the diagrammatic language of number and magnitude. This tendency is especially evident in Du Puy's (1612) diagram for X.5. There, in addition to representing the numerical quantities, the Hindu–Arabic numerals also appear next to the magnitudes, indicating the number of equal parts each line contains. Rather than preserving the traditional distinction between arithmetical and geometrical notation, such choices reveal a growing fluidity in how early modern editors visualized quantity.

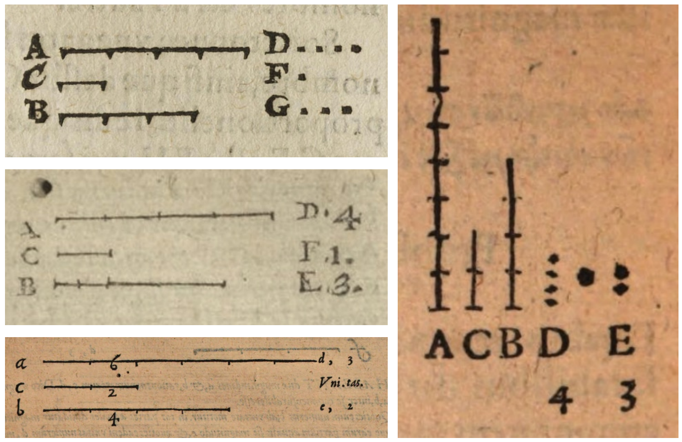


Figure 10: Diagrams for Proposition X.5. The upper left diagram (Dounot, 1610) represents numbers only with dots. The upper right diagram (Magnien and Gracilis, 1564) pairs the dots with Hindu–Arabic numerals. The middle row (Henrion, 1615) uses Hindu–Arabic numerals alone. The lower row (Du Puy, 1612) uses Hindu–Arabic numerals both for the numbers and for indicating the magnitudes of the lines.

7. Toward an explanation

As this paper approaches its end, a *why* question begins to surface. This study examined *how* the visual language of quantities evolved in early modern editions of Euclid's *Elements*. The question of *why*, that is, why these visual syntaxes and semantics changed, remains beyond the present scope. In many ways, the *why* is built upon the *how*: understanding the reasons behind the changes first requires establishing how those changes occurred. Yet, before concluding, it seems worthwhile to pause and briefly consider some aspects of the *why* question.

Previous studies point to several factors that may explain the changes in the concept of quantity and its visual representation. First, the mathematical books published during this period, including the *Elements*, were not necessarily aimed at the same readership. Later editions that were examined might have been designed for audiences more interested in the mathematical and didactic aspects of the text than in the restoration of the Greek original (Høyrup, 2019). This trend is also related to the expansion of mathematical education at that time, both within and beyond universities (Cifoletti, 1992). In parallel, abacus cultures gained significance, influencing a growing class of merchants and artisans and giving rise to a proliferation of abacus treatises and their conventions (Heeffer, 2022). At the same time, the rise of symbolic mathematics in scholarly circles likely also played a role (Heeffer and Van Dyck, 2010). Thus, the transformation of the concept of quantity seems to interact with a dynamic and multi-layered intellectual and educational landscape.

From a more concrete perspective, several centers and figures appear to have shaped the adoption of this visual language and its associated strategies. This study highlights some of them. First, in the mid-sixteenth century, the editions of Xylander and Scheubel published in Basel introduced notable innovations to the visual language of the *Elements*. Their works, together with those of other cossists, had a considerable impact on contemporary mathematical practice and likely contributed indirectly to the diffusion of these techniques (Loget, 2012). Another central figure emerging from this study is Clavius. His extensive use of both dots and Hindu–Arabic numerals and fractions in the diagrams of the arithmetical books and in Book X, as well as the ubiquitous use of dotted lines in the geometrical books marked a significant shift. Yet perhaps even more consequential than the visual innovations was the scale of his influence. As a leading figure within the Jesuit educational network and a prominent scholar, he occupied a position that enabled a wide dissemination of his pedagogical approaches.

Still, these shifts cannot be attributed to a small number of centers or individual figures. Instead, as this study has shown, the emergence of this visual language appears to stem from an accumulation of minor innovations dispersed across editions, reprints, and translations, which together signal a broader trend. Moreover, the diagrams that ultimately reached the readers do not straightforwardly reflect authorial intention. They are the outcome of a collective production process involving editors, printers, engravers, and publishers, and they are shaped by material and practical constraints of print. These conditions varied significantly across contexts. Each edition, and even some of the reprints, constituted a unique configuration, defined by specific relationships between editors and printers, local norms, institutional settings, intended circulation, and material and production choices. In the case of Clavius, for example, correspondence with his contemporaries suggests that publishers sometimes printed his work carelessly or misrepresented his intentions, and that the timing and content of his publications were shaped by the publication of competing books, market forces, scholarly debates, and political reasons (Clavius, 1992, esp. letters 36, 72, 114 and 178). Therefore, the *why* question necessarily encompasses a wider constellation of factors. These include deliberate pedagogical choices and contingent decisions, institutional frameworks, and economic pressures, all interacting over time in a feedback process that gradually shaped readers' horizons of expectation.

Lastly, to address the *why* question more fully, attention must shift beyond the Euclidean tradition. The flourishing of printed textbooks during this period brought forth a wide range of visual strategies. A preliminary survey of contemporary mathematical works shows that dotted lines and related techniques involving dots were by no means unique to the Euclidean texts (see Figure 11). The dotted lines found in the *Elements*' arithmetical books can be linked to earlier arithmetic traditions, from Boethius's *De arithmetica* manuscripts and its incunable editions to other early textbooks such as Luca Pacioli's *Summa de arithmetica*.[23] Similar devices appear in elementary manuals that taught basic operations with counters, primarily aimed at merchants.[24] Diagrams featuring scaled rulers, whether in general contexts of measurement or in the domain of military engineering, offer another example.[25] Practical and rudimentary geometry textbooks also employed dotted lines well before their integration into the geometrical books of Euclid's *Elements*.[26] Books related to surveying, perspective, cartography, architecture, gnomonics, as well as astronomy, mechanics and advanced mathematics incorporated dotted lines in their diagrams as well, often introducing complex visual language that combines different types of lines.[27]

This brief and partial survey serves to emphasize that tracing the practice of dotted lines and addressing the *why* question in a meaningful way requires situating the *Elements* within the broader visual, scholarly, and pedagogical cultures of the early modern period. The editions of the *Elements* did not emerge in isolation, and the playful experimentation with diagrammatic

[23] The citations here and in the following footnotes appear according to their order in Figure 11 (clockwise from upper left): Boethius, f. 41v; Boethius (1488, cap. 7); Pacioli (1494, f. B7v/f.15v).

[24] Böschenstein (1520, f. 4v); Schreiber (1521).

[25] Lanteri (1557, p. 77); Peverone (1558, p. 120).

[26] Mydorge (1630, p. 57); Cataneo (1584, f. 27r); Pomodoro and Scala (1599, plate 3); Schmid (1539, p. 22).

[27] As examples of early modern works that integrate diagrams with dotted lines across disciplines, see for surveying: Vinet (1577); for gnomonics: Clavius (1581); for geography: Apian and Frisius (1574); for astronomy: Kepler (1609); for mathematical treatises: Rivault (1615); for architecture and perspective: De l'Orme (1626) and in the context of military architecture: Marolois (1628); Bachot (1598); for basic mechanics: Mögling (1629); and in the context of instruments: Errard (1584).

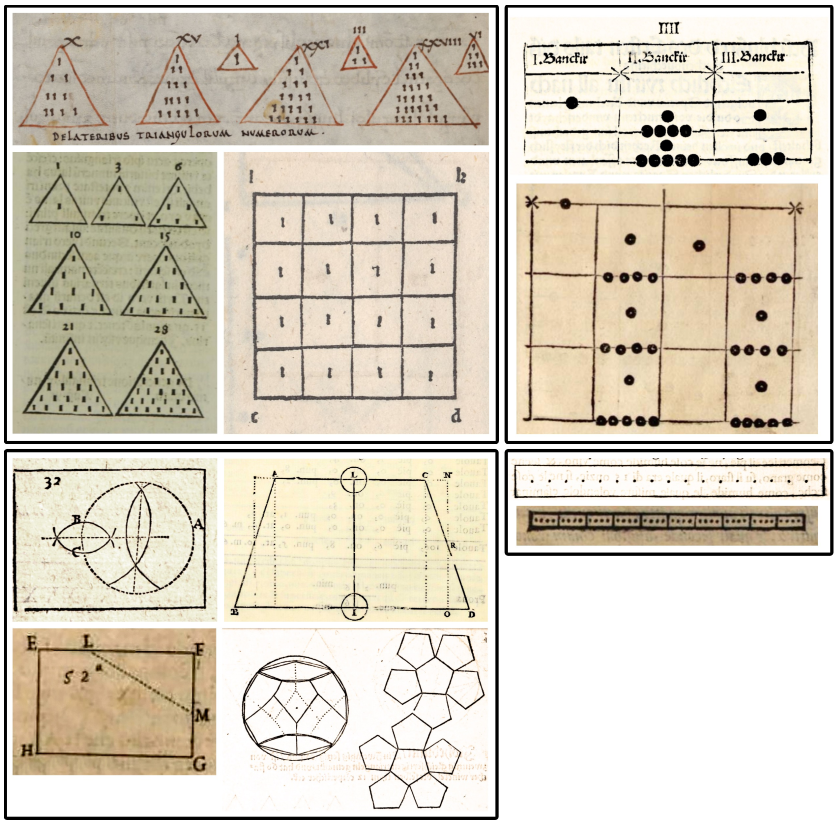


Figure 11: Representations related to dotted lines from various mathematical books. Upper left: triangular and square numbers in early arithmetic, from a tenth-century manuscript of Boethius's *De arithmetica*, an incunable edition, and Pacioli's *Summa de arithmetica*. Upper right: diagrams from sixteenth-century textbooks teaching how to use counters. Bottom left: examples from recreational, practical, and rudimentary geometry books from the late sixteenth and early seventeenth centuries. Bottom right: scale ruler diagrams in sixteenth-century military and practical textbooks, appearing in the context of measurement.

techniques characteristic of the time formed an essential part of its visual environment.

8. Conclusions

This paper set out to explore how the evolving visual language of dotted lines in early modern editions of Euclid's *Elements* contributed to the reconfiguration of the concept of quantity. The findings show that dotted lines have fulfilled two distinct roles: an arithmetical one, representing discrete multitudes of units, and a geometrical one, representing continuous magnitudes. In the arithmetical books, where dots stood for specific, countable units, this practice encountered difficulties when dealing with large numbers and impossible cases. Editors adopted a range of techniques to address these challenges and increasingly relied on Hindu–Arabic numerals, a shift that supported the representation of relations between numbers and promoted a more abstract conception of number. One culmination of this transition can be seen in Clavius's second edition of the *Elements* (1589), where fractional numbers replaced dotted configurations in the scholia section, mirroring the original diagrammatic structure while charging it with a new conception of number. By contrast, in the geometrical books, dotted lines responded to a different set of demands. They were used to represent magnitudes, for example, in auxiliary constructions or to mark hypothetical steps in proofs. While appearing only sporadically in sixteenth-century editions, dotted lines became increasingly widespread and conventional during the first decades of the seventeenth century.

The paper further proposes that the blurring boundary between the arithmetic and geometric domains is reflected in the visual representation of dotted lines. Rhode's (1609) edition, which applied similar visual forms across the arithmetical and geometrical books, as well as some renderings of Book X, illustrates how the boundaries between the two conceptions of quantity became less sharply defined and suggests that a degree of visual convergence emerged between them. It also demonstrates how Hindu–Arabic numerals increasingly functioned as intermediary symbols, bridging diagrammatic practices associated with number and magnitude.

This study highlights the connection between editorial context and diagrammatic practice. It suggests that changes in the functions of dotted lines and in the visual representation of notions such as number, magnitude, and hypothetical cases cannot be attributed to a single center, figure, or tradition.

At the same time, it highlights the prominent role that certain editorial contexts played, most notably Clavius's editions and the Basel translations of the mid-sixteenth century. The findings further point to possible exchanges with other diagrammatic and notational traditions beyond the Euclidean corpus, opening avenues for further study.

Finally, while the primary contribution of this study is historical, it also offers a methodological framework that opens up new directions for research. To navigate a corpus of this scale, a machine-learning system was employed to automatically extract visual elements from digital facsimiles, enabling the efficient identification and classification of diagrams. This approach builds on decades of digitization initiatives and on the expanding development of digital humanities tools for the study of historical sources. In this respect, the study not only proposes a methodological model for approaching visual features in early modern texts, but also provides concrete open-source tools designed to support similar investigations. These tools, easily adaptable to other contexts, aim to support the growing field of digital humanities in the history of early modern books and the history of mathematics and science in particular. The methodology can facilitate further exploration of questions raised by this study and it can also be applied to explore other forms of visual vocabulary within and beyond the Euclidean corpus, including materials produced by less-studied groups such as students, navigators, surveyors and merchants.

9. Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used ChatGPT, a generative AI tool, to assist with language editing, phrasing, and improving the clarity and readability of the text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

References

Alexander-Skipnes, I. (Ed.), 2017. Visual Culture and Mathematics in the Early Modern Period. Routledge, New York. doi:10.4324/9781315563435.

Andersen, K., Bos, H., 2006. Pure Mathematics. Cambridge University Press. The Cambridge History of Science, p. 696–724.

Anonymous, 1538. Euclidis Elementa. Rihel, Strasbourg.

Anonymous, 1559. Euclidis Elementa. Rihel, Strasbourg.

Anonymous, 2026. Dataset accompanying the present study. Online repository. Available upon request; anonymized for peer review.

Apian, P., Frisius, G., 1574. Cosmographia. Heirs of Arnold Birckmann, Cologne.

Artmann, B., 1999. Euclid—The Creation of Mathematics. Springer, New York. doi:10.1007/978-1-4612-1412-0.

Axworthy, A., 2023. The hybridization of practical and theoretical geometry in the sixteenth-century euclidean tradition. Journal of Interdisciplinary History of Ideas 11. doi:10.13135/2280-8574/7333.

Bachot, A., 1598. Le gouvernail. Ambroise Bachot, Melun and Paris. URL: https://catalogue.bnf.fr/ark:/12148/cb30044864m. printed in Melun and also sold in Paris.

Baigrie, B.S. (Ed.), 1996. Picturing Knowledge: Historical and Philosophical Problems Concerning the Use of Art in Science. University of Toronto Press, Toronto. URL: https://www.jstor.org/stable/10.3138/9781442678477, doi:10.3138/9781442678477.

Baldasso, R., 2013. Printing for the doge: On the first quire of the first edition of the liber elementorum euclidis. La Bibliofilía 115, 525–552. URL: https://www.jstor.org/stable/10.2307/26202240.

Bender, J., Marrinan, M., 2010. The Culture of Diagram. Stanford University Press, Stanford.

Billingsley, H., Dee, J., 1570. The Elements of Geometrie of the Most Ancient Philosopher Euclide of Megara. Daye, London. URL: https://archive.org/details/McGillLibrary-rbsc_elements-geometrie-evclide-megara_folioQA31E867131570-21289.

Boethius, . De institutione arithmetica. Manuscript, Corpus Christi College, University of Cambridge. URL: https://parker.stanford.edu/parker/catalog/xb651hx1944. mid-10th century.

Boethius, 1488. De institutione arithmetica. Ratdolt, Augsburg. URL: https://archive.org/details/ita-bnc-in2-00001756-001.

Böschenstein, J., 1520. Ain neu geordnet rechenbüchlein. Oeglin and Miller, Augsburg. URL: https://data.onb.ac.at/rep/105D229B.

Bredekamp, H., 2019. Galileo's Thinking Hand: Mannerism, Anti-Mannerism, and the Virtue of Drawing in the Foundation of Early Modern Science. De Gruyter, Berlin, Boston. doi:doi:10.1515/9783110539219.

Büttner, J., Martinetz, J., El-Hajj, H., Valleriani, M., 2022. Cor deep and the sacrobosco dataset: Detection of visual elements in historical documents. Journal of Imaging 8, 285. doi:10.3390/jimaging8100285.

Caiani, A., 1545a. Euclidis Elementorum libri XV. Blado, Rome.

Caiani, A., 1545b. I quindici libri degli Elementi di Euclide, di greco tradotti in lingua thoscana. Blado, Roma.

Campano, 1482. Preclarissimus liber elementorum Euclidis perspicacissimi, in artem geometrie incipit quam foelicissime. Ratdolt, Venice.

Cataneo, G., 1584. Dell'arte del misurare libri due, nel primo de' quali s'insegna a misurare et partir i campi. Nel secondo a misurar le muraglie imbottar grani vini fieni & strami; col liuellar dell'acque & altre cose necessarie a gli agrimensori. Libri primo. Bozzola, Brescia. URL: https://www.proquest.com/docview/2090358011.

Chemla, K.C. (Ed.), 2012. History of Mathematical Proof in Ancient Traditions. Cambridge University Press, Cambridge.

Cifoletti, G.C., 1992. Mathematics and Rhetoric: Peletier and Gosselin and the Making of the French Algebraic Tradition. Ph.D. thesis. Princeton University.

Clavius, C., 1574. Euclidis elementorum libri XV. Accolto, Rome. URL: https://archive.org/details/euclidiselemento01eucl.

Clavius, C., 1581. Gnomonices libri octo. Zanetti, Rome.

Clavius, C., 1589. Euclidis elementorum libri XV. Grassi, Rome. URL: https://mdz-nbn-resolving.de/details:bsb11770103.

Clavius, C., 1591. Euclidis elementorum libri XV; accessit XVI de solidorum regularium cuiuslibet intra quodlibet comparatione. Ciotti, Cologne. URL: https://mdz-nbn-resolving.de/details:bsb11770081.

Clavius, C., 1655. Sex Primi Elementorum Geometricorum. Dominici Grialdi; Typis Nicolai Angeli Tinassii, Romae.

Clavius, C., 1992. Corrispondenza. Edizioni del Dipartimento di Matematica dell'Università di Pisa, Pisa.

Commandino, F., 1572. Euclidis elementorum libri XV una cum scholiis antiquis. Franceschini, Pesaro. URL: https://hdl.handle.net/2027/ucm.531653091x.

Commandino, F., Spacciuoli, V., 1575. De gli elementi d'Euclide libri quindici con gli scolii antichi. Frisolino, Urbino.

Commandino, F., Spacciuoli, V., 1619. De gli elementi d'Euclide libri quindici con gli scolii antichi. Concordia, Urbino.

Corry, L., 2015. A Brief History of Numbers. 1 ed., Oxford University Press, Oxford.

De Foix-Candale, F., 1566. Euclidis Megarensis mathematici clarissimi elementa geometrica libri XV. Royer, Paris.

De l'Orme, P., 1626. Œuvres de Philibert de l'Orme. Chaudière, Paris.

De Montdoré, P., 1551. Euclidis elementorum liber decimus. Vascosan, Paris.

De Risi, V., 2016. The development of euclidean axiomatics. Archive for History of Exact Sciences 70, 591–676. doi:10.1007/s00407-015-0173-9.

De Young, G., 2005. Diagrams in the arabic euclidean tradition: A preliminary assessment. Historia Mathematica 32, 129–179. URL: https://www.sciencedirect.com/science/article/pii/S0315086004000424, doi:10.1016/j.hm.2004.04.003.

Dou, J.P., 1606. De ses eerste boecken Euclidis. Bouwensz., Leiden.

Dounot, D., 1610. Les éléments de la géométrie d'Euclide Mégarien. Jacques le Roy, Paris. URL: https://dlc.mpg.de/image/868448656/7, doi:10.48644/868448656.

Dounot, D., 1613. Les éléments de la géométrie d'Euclide Mégarien. Jacques le Roy, Paris.

Du Puy, F., 1612. Euclidis elementorum decimum. de Heuqueville, Paris. URL: https://mdz-nbn-resolving.de/details:bsb11203766.

Dyvad, C., 1603. In geometriam Euclidis prioribus sex elementorum libris comprehensam demonstratio linealis. Jansz, Arnhem.

Errard, J., 1584. Le premier livre des instruments mathematiques mechaniques. Jan Janson, Nancy. URL: http://catalogue.bnf.fr/ark:/12148/cb437046508.

Errard, J., 1598. Les six premiers livres des Éléments d'Euclide. Auvray, Paris.

Faietti, M., Wolf, G. (Eds.), 2015. The Power of Line. Hirmer Publishers, Munich.

Fine, O., 1536. In sex priores libros geometricorum elementorum Euclidis Megarensis demonstrationes. de Colines, Paris.

Fine, O., 1544. In sex priores libros geometricorum elementorum Euclidis Megarensis demonstrationes. de Colines, Paris.

Fine, O., 1551. In sex priores libros geometricorum elementorum Euclidis Megarensis demonstrationes. Chaudière, Paris.

Forcadel, P., 1564. Les six premiers livres des Éléments d'Euclide, traduits et commentés. Marnef, Paris.

Forcadel, P., 1565. Les septième, huitième et neuvième livres des Éléments d'Euclide. Perier, Paris.

Forcadel, P., 1566. Les six premiers livres des Éléments d'Euclide. Perier, Paris.

Fowler, D.H., 1992. An invitation to read book x of euclid's elements. Historia Mathematica 19, 233–264. doi:10.1016/0315-0860(92)90029-B.

Gestrinius, M., 1642. In geometriam Euclidis demonstrationum libri sex. Matthiae, Uppsala.

Goldstein, J.A., 2000. A matter of great magnitude: The conflict over arithmetization in 16th-, 17th-, and 18th-century english editions of euclid's elements books i through vi (1561–1795). Historia Mathematica 27, 36–53. URL: https://www.sciencedirect.com/science/article/pii/S0315086099922632, doi:10.1006/hmat.1999.2263.

Grynaeus, S., 1533. Euclidis Elementa. Herwagen, Basel.

Hamami, Y., Mumma, J., 2013. Prolegomena to a cognitive investigation of euclidean diagrammatic reasoning. Journal of Logic, Language and Information 22, 421–448. doi:10.1007/s10849-013-9182-8.

Heath, T.L., 1908. The Thirteen Books of Euclid's Elements: Translated from the Text of Heiberg, with Introduction and Commentary. Cambridge University Press, Cambridge.

Heeffer, A., 2022. Arithmetic in the renaissance, in: Sgarbi, M. (Ed.), Encyclopedia of Renaissance Philosophy. Springer, Cham. doi:10.1007/978-3-319-14169-5_1174.

Heeffer, A., Van Dyck, M., 2010. Philosophical Aspects of Symbolic Reasoning in Early Modern Mathematics. College Publications, London.

Henrion, D., 1615. Les quinze livres des Éléments d'Euclide, traduits du latin en français. Joallin, Paris. URL: https://gallica.bnf.fr/ark:/12148/bpt6k8702015c.

Henrion, D., 1621. Les quinze livres des Éléments d'Euclide, traduits du latin en français. Joallin, Paris.

Henrion, D., 1632. Les quinze livres des Éléments d'Euclide, traduits du latin en français. Dedin, Paris. URL: https://gallica.bnf.fr/ark:/12148/bpt6k1510937s.

Herlin, C., 1537. Euclidis Megarensis mathematici clarissimi elementorum geometricorum libri XV. Herwagen, Basel. URL: `https://mdz-nbn-resolving.de/details:bsb11724610`.

Herlin, C., 1546. Euclidis Megarensis mathematici clarissimi elementorum geometricorum libri XV. Herwagen, Basel.

Herlin, C., 1559. Euclidis Megarensis mathematici clarissimi elementorum geometricorum libri XV. Herwagen and Brand, Basel.

Høyrup, J., 2017. What is "geometric algebra," and what has it been in historiography? AIMS Mathematics 2, 128–160. doi:`10.3934/Math.2017.1.128`.

Høyrup, J., 2019. Euclid: Reception in the renaissance, in: Sgarbi, M. (Ed.), Encyclopedia of Renaissance Philosophy. Springer International Publishing. doi:`10.1007/978-3-319-02848-4_918-1`.

Jones, C.A., Galison, P. (Eds.), 1998. Picturing Science, Producing Art. Routledge, New York.

Kaiser, D., 2005. Drawing Theories Apart: The Dispersion of Feynman Diagrams in Postwar Physics. University of Chicago Press, Chicago.

Katz, V.J., 2009. A History of Mathematics: An Introduction. Third ed., Pearson, Boston.

Kepler, J., 1609. Astronomia Nova. Vögelin, Heidelberg.

Lanteri, G., 1557. Due dialoghi di m. Iacomo de' Lanteri da Paratico bresciano; ne i quali s'introduce messer Girolamo Catanio novarese & messer Francesco Trevisi ingegnero veronese con un giovene bresciano a ragionare del modo di disegnare le piante delle fortezze secondo Euclide; et del modo di comporre i modelli et torre in disegno le piante della città. Valgrisi & Costantini, Venezia. URL: `http://id.sbn.it/bid/BVEE018640`.

Lanz, J., 1617. Euclidis elementorum geometricorum libri sex priores. Angermayer, Ingolstadt.

Lanz, J., 1629. Euclidis elementorum geometricorum libri sex priores. Rossi, Bologna.

Lee, E., 2018. Let the diagram speak: Compass arcs and visual auxiliaries in printed diagrams of euclid's elements. Endeavour 42, 78–98. doi:10.1016/j.endeavour.2018.07.010.

Lee, E., 2020. Visual Agency in Euclid's Elements: A Study of the Transmission of Visual Knowledge. Ph.D. thesis. Stanford University.

Lefèvre, J., 1516. Euclidis Megarensis geometricorum elementorum libri XV. Estienne, Paris. URL: https://mdz-nbn-resolving.de/details:bsb10195529. another digitized facsimlie available at https://mdz-nbn-resolving.de/details:bsb11111685.

Loget, F., 2012. Printers and algebraists in mid-16th century france. Philosophica 87, 83–116. doi:10.21825/philosophica.82140.

Macbeth, D., 2010. Diagrammatic reasoning in euclid's elements, in: van Kerkhove, B., de Vuyst, J., van Bendegem, P. (Eds.), Philosophical Perspectives on Mathematical Practice. College Publications, London, pp. 235–267.

Magnien, J., Gracilis, S., 1564. Euclidis elementorum libri XV, Graece et Latine. Cholinus, Cologne. URL: https://mdz-nbn-resolving.de/details:bsb10994218.

Malet, A., 2006. Renaissance notions of number and magnitude. Historia Mathematica 33, 63–81. doi:10.1016/j.hm.2004.11.011.

Manders, K., 2008. The euclidean diagram, in: Mancosu, P. (Ed.), The Philosophy of Mathematical Practice. Oxford University Press, Oxford, pp. 65–84. doi:10.1093/acprof:oso/9780199296453.003.0005.

Marius, S., 1610. Die ersten sechs Bücher Elementorum Euclidis. Böhm, Ansbach.

Marolois, S., 1628. Fortification ou Architecture militaire, tant offensive que defensive, supputée et designée. Guillaume Jansson Caesius, Amsterdam.

Maurolico, F., 1575. Opuscula mathematica. Franceschi, Venice.

Mögling, D., 1629. Mechanischer Künste Kammer. Erster Theil, begreiffend die wahre fundamenta aller Machination. Rotel, for Merian, Frankfurt am Main.

Mueller, I., 1981. Philosophy of Mathematics and Deductive Structure in Euclid's Elements. Dover Publications, Mineola, NY.

Mumma, J., Panza, M., Sandu, G. (Eds.), 2012. Diagrams in Mathematics: History and Philosophy. volume 186. Synthese. Special issue, May 2012.

Mydorge, C., 1630. Examen du livre des recreations mathematiques: et de ses problemes en geometrie, mechanique, optique, & catoptrique; ou sont aussi discutées & restablies plusieurs experiences physiques y proposées. volume 1. Boutonne, Paris. URL: https://data.onb.ac.at/rep/10ABEC20.

Naboth, V., 1556. Euclidis Megarensis mathematici clarissimi elementorum geometricorum liber primus. Birkmann, Köln.

Naets, J., 2010. How to define a number? a general epistemological account of simon stevin's art of defining. Topoi 29, 77–86. doi:10.1007/s11245-009-9068-1.

Nasifoglu, Y., 2021. Reading by drawing: The changing nature of mathematical diagrams in seventeenth-century england, in: Nasifoglu, Y., Wardhaugh, B., Beeley, P. (Eds.), Reading Mathematics in Early Modern Europe. Routledge, United Kingdom, pp. 62–101. doi:10.4324/9781003102557-4.

Neal, K., 2002. From Discrete to Continuous: The Broadening of Number Concepts in Early Modern England. Studies in History and Philosophy of Science, Springer, Dordrecht. doi:10.1007/978-94-017-0077-1.

Netz, R., 1999. The Shaping of Deduction in Greek Mathematics: A Study in Cognitive History. Cambridge University Press, Cambridge.

Oaks, J.A., 2018. François viète's revolution in algebra. Archive for History of Exact Sciences 72, 245–302. doi:10.1007/s00407-018-0208-0.

Oosterhoff, R., 2018. Making Mathematical Culture: University and Print in the Circle of Lefèvre d'Étaples. Oxford-Warburg Studies, Oxford University Press, Oxford.

Pacioli, L., 1494. Somma di arithmetica, geometria, proporzioni e proporzionalità. Paganini, Venice. URL: https://repository.teneo.libis.be/delivery/DeliveryManagerServlet?dps_pid=IE6857464.

Pacioli, L., 1509. Euclidis megarensis philosophi acutissimi mathematicorumque omnium sine controversia principis opera. Paganino, Venice.

Pantin, I., 2022. Mathematical books in paris (1531–1563): The development of publishing strategies in a competitive international market, in: Valleriani, M., Ottone, A. (Eds.), Publishing Sacrobosco's De sphaera in Early Modern Europe: Modes of Material and Scientific Exchange. Springer International Publishing, Cham, pp. 289–335. doi:10.1007/978-3-030-86600-6_9.

Pauwels, L., 2006. Visual Cultures of Science: Rethinking Representational Practices in Knowledge Building and Science Communication. Dartmouth College Press, Hanover.

Peletier, J., 1557. In Euclidis Elementa geometrica demonstrationum libri sex. Tornes, Lyon.

Peletier, J., 1610. In Euclidis elementa geometrica demonstrationum libri sex. de Tournes, Geneva.

Peletier, J., 1628. Les six premiers livres des éléments géométriques d'Euclide. de Tournes, Geneva.

Peverone, G.F., 1558. Due breui e facili trattati, il primo d'arithmetica, l'altro de geometria, ne i quali si contengono alcune cose nuoue piaceuoli e utili si a gentilhuomini come artegiani. di Tournes, Lyon. URL: https://mdz-nbn-resolving.de/details:bsb10525716.

Pomodoro, G., Scala, G., 1599. Geometria prattica tratta dagl'Elementi d'Euclide et altri auttori da Giovanni Pomodoro venetiano mathematico eccellentissimo descritta et dichiarata da Giovanni Scala mathematico. Paolini, Roma. URL: https://archive.org/details/bub_gb_ZB7tcNVx27cC.

Priest, G., De Toffoli, S., Findlen, P., 2018. Tools of reason: The practice of scientific diagramming from antiquity to the present. Endeavour 42, 49–59. URL: https://www.sciencedirect.com/science/article/pii/S0160932718300504, doi:10.1016/j.endeavour.2018.07.001.

De la Ramée, P., 1549. Euclides. Richard, Paris.

De la Ramée, P., 1558. Euclides. Richard, Paris.

Rhäticus, G.J., Camerarius, J., 1549. Euclidis elementorum geometricorum libri sex, conversi in Latinum sermonem. Valentin, Leipzig.

Rhode, A., 1609. Euclidis elementorum libri XIII. Helwig, Wittenberg. URL: https://mdz-nbn-resolving.de/details:bsb10235952.

Rhode, A., 1634. Euclidis elementorum libri XIII. Helwig, Wittenberg.

Rivault, D., 1615. Archimedis opera quae extant, novis demonstrationibus commentariisque illustrata. Morel, Paris.

Rommevaux, S., 2006. Clavius: Une clé pour Euclide au XVIe siècle. Vrin, Paris.

Rudd, T., 1651. Euclides: Elements of Geometry. Robert and William Leybourn for Richard Tomlins and Robert Boydell, London. URL: https://archive.org/details/b30337070.

Saito, K., 2018. Diagrams and traces of oral teaching in euclid's elements: Labels and references. ZDM Mathematics Education 50, 921–936. doi:10.1007/s11858-018-0929-1.

Saito, K., 2019. Re-examination of the different origins of the arithmetical books of euclid's elements. Historia Mathematica 47, 39–53. doi:10.1016/j.hm.2019.03.002.

Saito, K., Sidoli, N., 2012. Diagrams and arguments in ancient greek mathematics: Lessons drawn from comparisons of the manuscript diagrams with those in modern critical editions, in: Chemla, K. (Ed.), The History of Mathematical Proof in Ancient Traditions. Cambridge University Press, Cambridge, pp. 135–162.

Schäffner, W., 2005. The point: The smallest venue of knowledge in the 17th century (1585–1665), in: Schramm, H., Schwarte, L., Lazardzig, J. (Eds.), Collection – Laboratory – Theater: Scenes of Knowledge in the 17th Century. De Gruyter, Berlin and New York. volume 1, pp. 57–74. doi:10.1515/9783110201550.57.

Scheubel, J., 1550. Euclidis Megarensis philosophi et mathematici excellentissimi sex libri priores de geometricis principiis. Herwagen, Basel.

Scheubel, J., 1555. Das sibend, acht und neünt büch des hochberümbten Mathematici Euclidis Megarensis. Valentin Ottmar, Augsburg.

Schmid, W., 1539. Das erst buch der geometria. Ein kurtze unterweisung was und warauff geometria gegruendet sey und wie man allerley lini flech und coerper außtheylen und in fuergegebner proportion machen soll. Peterlein, Nuremberg. URL: https://mdz-nbn-resolving.de/details:bsb10164586.

Schreiber, H., 1521. Ayn new kunstlich Buech welches gar gewiß vnd behend lernet nach der gemainen regel Detre, welschen practic, regeln falsi vnd etlichen regeln Cosse, mancherlay schöne vnd zuwissen notürfftig rechnung auff kauffmanschafft. Johann Stuchs, Nürnberg. URL: http://mdz-nbn-resolving.de/urn:nbn:de:bvb:12-bsb00028744-3. printed for Lukas Alantsee, bookseller in Vienna.

Schubring, G., 2025. Mathematics textbooks: Transitions from manuscript to print. ZDM Mathematics Education 57, 963–986. doi:10.1007/s11858-025-01709-9.

Steensen, A.K., Johansen, M.W., 2016. The role of diagram materiality in mathematics. Cognitive Semiotics 9, 183–201. doi:10.1515/cogsem-2016-0008.

Stevin, S., d'Alexandrie, D., 1585. L'Arithmétique. Plantin, Leyde. URL: https://gallica.bnf.fr/ark:/12148/bpt6k1521833c.

Sthen, J., 1564. Arithmetices Euclideae liber primus. Wittenberg.

Tartaglia, N., 1543. Euclide Megarense philosopho, solo introduttore delle scientie mathematice. Ruffinelli, Venice.

Tartaglia, N., 1565. Euclide Megarense philosopho, solo introduttore delle scientie mathematice. Curtio Troiano, Venice.

Tartaglia, N., 1569. Euclide Megarense, acutissimo filosofo, solo introduttore delle scientie mathematiche. Giovanni Bariletto, Venice.

Tartaglia, N., 1585. Euclide Megarense philosopho, solo introduttore delle scientie mathematice. The Heirs of Curtio Troiano, Venice.

Tartaglia, N., 1586. Euclide Megarense philosopho, solo introduttore delle scientie mathematice. The Heirs of Curtio Troiano, Venice.

Vinet, É., 1577. L'Arpanterie d'Élie Vinet, livre de géométrie, enseignant à mezurer les champs, et pluzieurs autres chozes. S. Millanges, Bourdeaus. URL: http://catalogue.bnf.fr/ark:/12148/cb31588660g.

Wagner, R., Netz, R., 2023. Between music and geometry: A proposal for the early intended application of euclid's elements book x. British Journal for the History of Mathematics 38, 69–96. doi:10.1080/26375451.2023.2197351.

Wardhaugh, B., Beeley, P., Nasifoglu, Y., 2020. Euclid in Print, 1482–1703: A Catalogue of the Editions of the Elements and Other Euclidean Works. The Bibliographical Society. URL: https://doi.org/10.5281/zenodo.4269490, doi:10.5281/zenodo.4269490.

Xylander, W., 1562. Die sechs ersten Bücher Euclidis, vom Anfang oder Grund der Geometrie. Kündig, Basel. URL: https://mdz-nbn-resolving.de/details:bsb11057832.

Zamberti, B., 1505. Euclidis megarensis philosophi platonici mathematicarum disciplinarum janitoris. Tacuino, Venice.

Zamberti, B., 1510. Euclidis megarensis philosophi platonici mathematicarum disciplinarum janitoris. Tacuino, Venice.

Zamorano, R., 1576. Los seis libros primeros de la geometría de Euclides. La Barrera, Seville.

Figure Captions

List of Figures

List of Tables